Artwork Similarity Search Report

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Approach:

Before coding, a proper study is conducted on the dataset being used and understanding of the underlying models and methodologies is revisited.

Baseline: DINOv2 Embeddings, a self-supervised Vision Transformer pretrained on millions of images using knowledge distillation is used. For each artwork, the CLS token of the final hidden layer is extracted (dimension = 384), representing global embeddings of the image. These extracted embeddings are stored in Qdrant VectorDB (on disk)

Hybrid Embedding: The first enhancement of the baseline model is done by combining DINOv2 embeddings with CLIP(ViT-B/32) via normalization and concatenation. Normalization is done to ensure equal contribution.The main motivation was that DINO captured visual features and CLIP captured conceptual meaning due to language supervision.

Reranker: CLIP similarity was chosen to rerank the top_k results from the baseline model. The main idea behind to choose this is that CLIP is already used in the hybrid embedding method. This creates a direct comparison between two ways of combining DINOv2 and CLIP, one as joint hybrid embeddings or as a sequential reranker. This allows for a evaluation of which combination strategy is better for artwork retrieval.

Top-10 candidates for the query image is returned by the baseline and then the results are ranked based on CLIP score between the query and each candidate to finally return the top-5 best matches.

Multi-View Aggregation: For the Met Museum dataset, multiple views of the same artwork are available. So all N views are embedded individually, normalised and average pooled into a single representation.

Trade-offs:

1. DInov2 Small was chosen over the Base model, mostly due to hardware constraint (RTX 3050, 4GB VRAM). This was done because of my initial approach to tackle hybrid embedding with VGG 16 feature maps through Gram Matrix.The main trade-off is smaller embedding dimension and lower retrieval accuracy.
2. The initial plan to use Gram Matrix features with VGG16 to capture the style of the artwork was dropped due to complexity and time constraint. Infact, a paper was found during study called FreestyleRet (arXiv:2312.02428v2) showed this approach, however heavier computation led to simpler CLIP concatenation.
3. Qdrant local removed docker dependency, however this did not allow for concurrent access to the same db.
4. Evaluation set for Met Museum was limited to small number due to the time constraint on manual image selection and downloads. This makes the quantitative results not a good conclusion statistically rather a indicator

What Worked:

1. DInoV2 performed well out of the box on the WikiArt retrieval task. The CLS token provided a strong signal.
2. CLIP concatenation worked and produced measurable improvement over DINOv2
3. CLIP reranking also improved on metrics like P@5 over DINOv2

What did not work:

1. CLIP reranking did not improve the TOP-1 accuracy.
2. Gram Matrix Hybrid experimentation failed, though it sounded good but was practically complex to carry out.
3. Evaluation of the Met dataset on the small evaluation dataset.

What would I improve If I had Time

1. More balanced and proper evaluation set for the Met museum dataset.
2. Gram Matrix and DINO V2 hybrid embeddings.
3. Improve the Reranking Top-1 accuracy.